



interfaceNode - *Datasheet*

Version 1.0.1

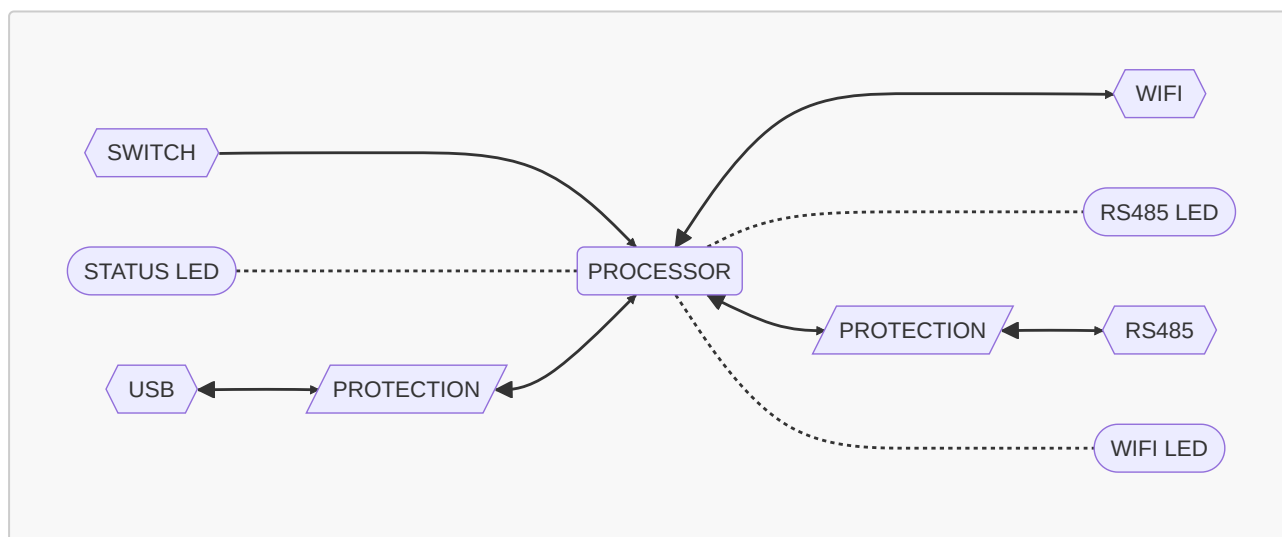
The Atkin Engineering **interfaceNode** is a Wireless RS-485 bridge that provides bridging connections between RS-485, USB, TCP sockets, and WebSockets for various control devices.

front

rear



Block Diagram



Specifications

Mechanical

- Dimensions 120 x 50 x 30 mm (4.72 x 2.36 x 1.18 in)
- Weight 105 grams (2.71 oz)

Electrical

- 12/24 VDC (absolute range 5 - 42 VDC) from RJ45 port
- 80mW average, range 25mW - 3W(peak) [[^]current]

Protections

Electrical

- AEC-Q100 Qualified
- Input Protection to +/- 42 VDC

RS485 Data

- IEC 61000-4-2 electrostatic discharge (ESD), Level 4 - 8k V / 15 kV (Contact / Air)
- IEC 61000-4-4 electrical fast transients (EFT), Level 4 - 2 kV
- IEC 61000-4-5 surge immunity, Level 2 - 4 kV
- ±70 VDC bus fault

USB

- Native USB CDC
- Isolation voltage 2.5kV
- Transient / Surge 5kV

Services

RS485

- Full / Half duplex (hardware jumper - see [[^]jumpers])
- Termination **None** / **120ohm** (hardware jumper - see [[^]jumpers])
- Protocol aware (DyNet 1 & 2, or raw passthrough)
- Baud rates: 9600 / 19200 / 38400 / 57600 / 115200

WiFi (Access Point / Station)

- IEEE 802.11b/g/n-compliant
- Supports 20 MHz and 40 MHz bandwidth in 2.4 GHz band
- 1T1R mode with data rate up to 150 Mbps
- Center frequency range of operating channel: 2412 ~ 2484 MHz
- WPA2/WPA3 Enterprise (802.1X) supported in Station mode

USB

- USB serial data via CDC, USB 2.0 full-speed

Bridge

- Bridges data between four interfaces: USB CDC, RS485, TCP sockets, and WebSockets
- Protocol aware (DyNet 1 & 2, checksum-validated) or raw passthrough - applies across all four interfaces
- WebSocket bridge traffic can be filtered by DyNet frame type (DyNet 1, DyNet 2, or both)

Connections

Front

- USB Type-C, USB 2.0 full-speed port [^{^usbpower}]
- 1x Status LED
- 1x System button

Rear

- 2x RS-485 (parallel RJ12 6P6 ports) [^{^rs485power}]
- 2x RGB status LED's

Environment

- Operating Temperature: 5° to 40°C ambient (41° to 104°F)
- Storage/Transport Temperature -20° to 65°C ambient (-4° to 149°F)
- Humidity 0 to 90% non-condensing

Compliance

- RF certification: CE, FCC, IC, SRRC, TELEC & BQB [^{^certification}]
- Green certification: REACH/RoHS

Supplied Accessories

- 1x USB Type-C to USB Type-A cable
- 1x RJ12 cable
- 1x Dual RJ12 to pin adapter

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[^{^current}]: Current requirements dependant on operating modes and environment.

[^{^usbpower}]: USB Type-C port does not supply power.

[^{^rs485power}]: RS-485 (parallel RJ12 6P6 ports) supplies power & data to **interfaceNode**.

[^{^jumpers}]: Termination and duplex are both set via jumpers internal to the enclosure, not configurable from the web interface - changing either requires opening the unit. For duplex specifically: the jumper selects between the receiver always being enabled (full duplex) or tied to the transmit-enable line (half duplex). The firmware drives a single direction-control signal in either case - the jumper, not the firmware, determines which duplex mode results.

[^{^certification}]: Final certification to be completed.

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